

Functional Time and Recoverability Window:

A Minimal Dynamical Framework for Endogenous Reachability Collapse (ERC)

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AI-assisted modeling and analysis using ChatGPT and Grok

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1. Abstract
2. Introduction
3. Minimal Recoverability Model
4. Functional Time and Recoverability
5. Ratio Organization
6. Sequential Perturbations and History Dependence
7. Stochastic Robustness
8. Numerical Robustness
9. Interpretation and Limitations
10. Conclusion
11. Appendix: Parameters and Reproducibility
12. References

1. Abstract

This exploratory computational note introduces a minimal recoverability-limited dynamical model to study whether operational accessibility can degrade before loss of dynamical state-space structure.

The model separates total perturbation load, perturbation delivery timescale, intrinsic recovery timescale, and available functional recovery window. Simulations show that recoverability is not determined by load alone, but by the relation between perturbation delivery and internal recovery dynamics. A recoverability boundary emerges in parameter space, functional recovery margin decreases continuously near this boundary, and the ratio between recovery timescale and perturbation timescale organizes the accessibility transition.

Sequential perturbation experiments further show history-dependent recoverability, where unresolved prior perturbations reduce later accessibility. Under stochastic forcing, the deterministic boundary broadens into a probabilistic transition band while preserving the same overall structure. Numerical sensitivity tests suggest that the main results are not artifacts of the chosen timestep or simulation horizon.

These results support an operational ERC framing: minimal systems may lose recoverable future accessibility before their underlying state-space structure disappears. The framework is operational and recoverability-focused rather than a proposal regarding fundamental physical time or spacetime geometry.

2. Introduction

Many dynamical systems remain structurally admissible while progressively losing the ability to recover from perturbation. In practice, systems often fail not because possible states disappear, but because return trajectories become operationally inaccessible within finite-time constraints. This distinction motivates a separation between dynamical admissibility and operational recoverability. A system may continue to possess accessible states in principle while simultaneously losing the capacity to resolve perturbations sufficiently fast to preserve recoverable behavior.

The present work explores this distinction using a minimal computational framework. Rather than modeling detailed physical or biological mechanisms, the goal is to investigate whether simple recoverability-limited systems naturally generate structured operational accessibility transitions when perturbation delivery and recovery dynamics interact across finite functional windows.

The model isolates four operational quantities:

- total perturbation load,
- perturbation delivery timescale,
- intrinsic recovery timescale,
- and available functional recovery time.

The central question is whether recoverability depends only on total perturbation magnitude, or whether the temporal relation between perturbation delivery and recovery dynamics also determines which trajectories remain operationally accessible.

Simulations show that recoverability is not governed by perturbation magnitude alone. Instead, operational accessibility depends strongly on the relation between perturbation delivery timescale and intrinsic recovery dynamics. The system exhibits structured recoverability boundaries, gradual degradation of functional recovery margin near transition regions, history-dependent recoverability under sequential perturbations, and probabilistic transition bands under stochastic forcing.

Importantly, the framework is operational rather than fundamental. The model does not propose modifications to physical time, relativity, or spacetime geometry. Instead, it explores how recoverability-limited systems may experience differing amounts of operationally accessible future despite sharing the same external elapsed duration.

Similar recoverability constraints may arise in biological, ecological, computational, or astrophysical systems where perturbation delivery competes with intrinsic recovery dynamics. The resulting framework may provide a useful minimal language for studying recoverability, accessibility collapse, and finite-time operational constraints across a broad class of dynamical systems.

3. Minimal Recoverability Model

To investigate operational recoverability under finite-time constraints, we constructed a minimal dynamical system subjected to transient perturbations and rate-limited recovery dynamics. The objective of the framework is not to reproduce a specific physical or biological mechanism, but to isolate the interaction between perturbation delivery and intrinsic recovery timescales within a controlled operational setting.

The system state variable is denoted by:

$v(t)$ and evolves under the combined effect of transient forcing and intrinsic relaxation dynamics.

Perturbations are modeled as finite-time Gaussian forcing functions:

$u(t)$

characterized by:

- total perturbation load (AUC),
- perturbation delivery timescale (σ),
- perturbation center time.

The Gaussian formulation allows perturbations with identical total load but different temporal delivery profiles.

Intrinsic recovery dynamics are controlled by a characteristic recovery timescale:

τ_r

which determines how rapidly the system relaxes back toward baseline after perturbation.

The model therefore separates:

1. perturbation magnitude,
2. perturbation delivery speed,
3. intrinsic recovery speed.

This separation is central to the framework.

Operational recoverability is evaluated within a finite functional recovery window:

T_{window}

rather than over infinite asymptotic time. A trajectory is classified as recoverable if the system returns below a predefined recovery threshold within the available operational window.

This operational criterion intentionally differs from classical asymptotic stability analysis. The focus is not whether recovery eventually occurs in principle, but whether recovery remains dynamically accessible within finite operational time constraints.

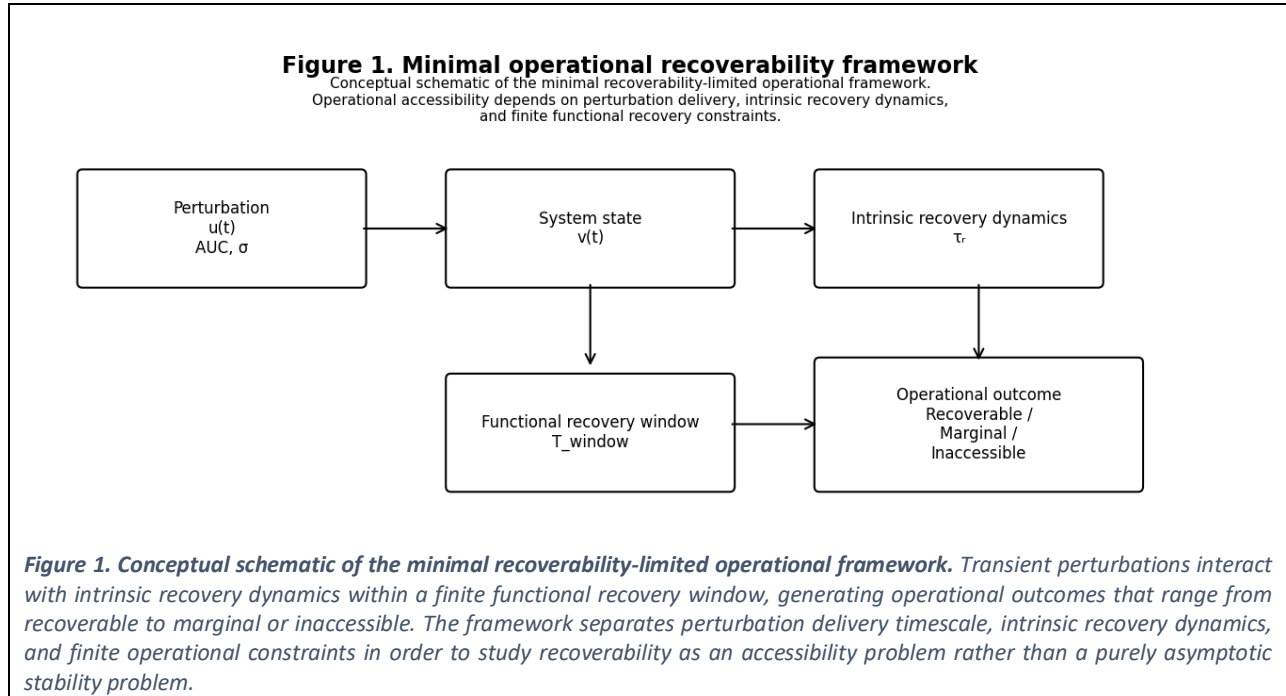
The primary quantities explored throughout the simulations are therefore:

- perturbation delivery timescale (σ),
- intrinsic recovery timescale (τ_r),
- functional recovery window (T_{window}),
- and the emergent ratio:

τ_r / σ

which later appears to organize the accessibility transition.

The framework was explored across deterministic, sequential, stochastic, and numerical robustness simulations in order to test whether the observed transitions persist under varying conditions.



4. Functional Time and Recoverability

4.1 Recoverability Boundary

The first set of simulations explored recoverability across a two-dimensional parameter space defined by:

- perturbation delivery timescale (σ),
- intrinsic recovery timescale (τ_r).

For each parameter combination, the system was evaluated according to the operational recoverability criterion described above.

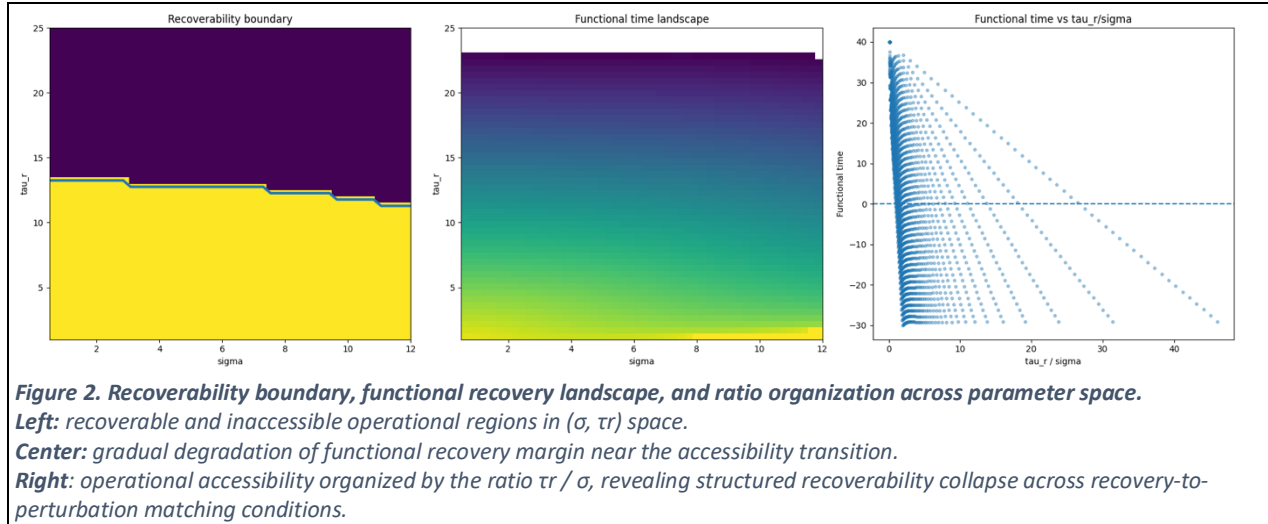
The simulations revealed the emergence of a structured recoverability boundary separating operationally accessible and inaccessible regions of parameter space.

Importantly, the transition was not determined solely by perturbation magnitude. Instead, recoverability depended strongly on the relation between perturbation delivery speed and intrinsic recovery dynamics.

Systems with sufficiently rapid recovery relative to perturbation delivery remained operationally recoverable within the available functional window. In contrast, systems with slower recovery

relative to perturbation delivery crossed a transition region beyond which operational return no longer occurred within finite-time constraints.

The resulting transition did not appear as a purely arbitrary threshold, but as a structured operational boundary emerging from the interaction between delivery and recovery timescales.



4.2 Functional Time Landscape

To further characterize the transition region, the simulations next evaluated the remaining functional recovery margin across parameter space.

Rather than classifying trajectories only as recoverable or non-recoverable, this analysis quantified how much operational recovery margin remained available before recoverability collapse occurred.

The resulting functional-time landscape revealed that accessibility degradation occurs progressively prior to complete loss of recoverability.

Near the transition region, systems retained progressively smaller operational recovery margins even while remaining technically recoverable according to the finite-time criterion.

This distinction is important because it suggests that recoverability collapse may emerge gradually through loss of operational accessibility before complete dynamical failure occurs.

The simulations therefore distinguish between:

- existence of dynamical trajectories,
- and operational accessibility of recoverable trajectories within finite functional time.

Importantly, all simulations shared the same external elapsed duration. The observed differences emerged from the interaction between perturbation delivery and intrinsic recovery dynamics rather than differences in external clock time itself.

5. Ratio Organization

The previous simulations suggested that recoverability was not controlled independently by perturbation delivery or recovery dynamics alone, but rather by their relative interaction.

To investigate this possibility, the simulations were reorganized according to the dimensionless ratio:

$$\tau_r / \sigma$$

where:

- τ_r represents intrinsic recovery timescale,
- σ represents perturbation delivery timescale.

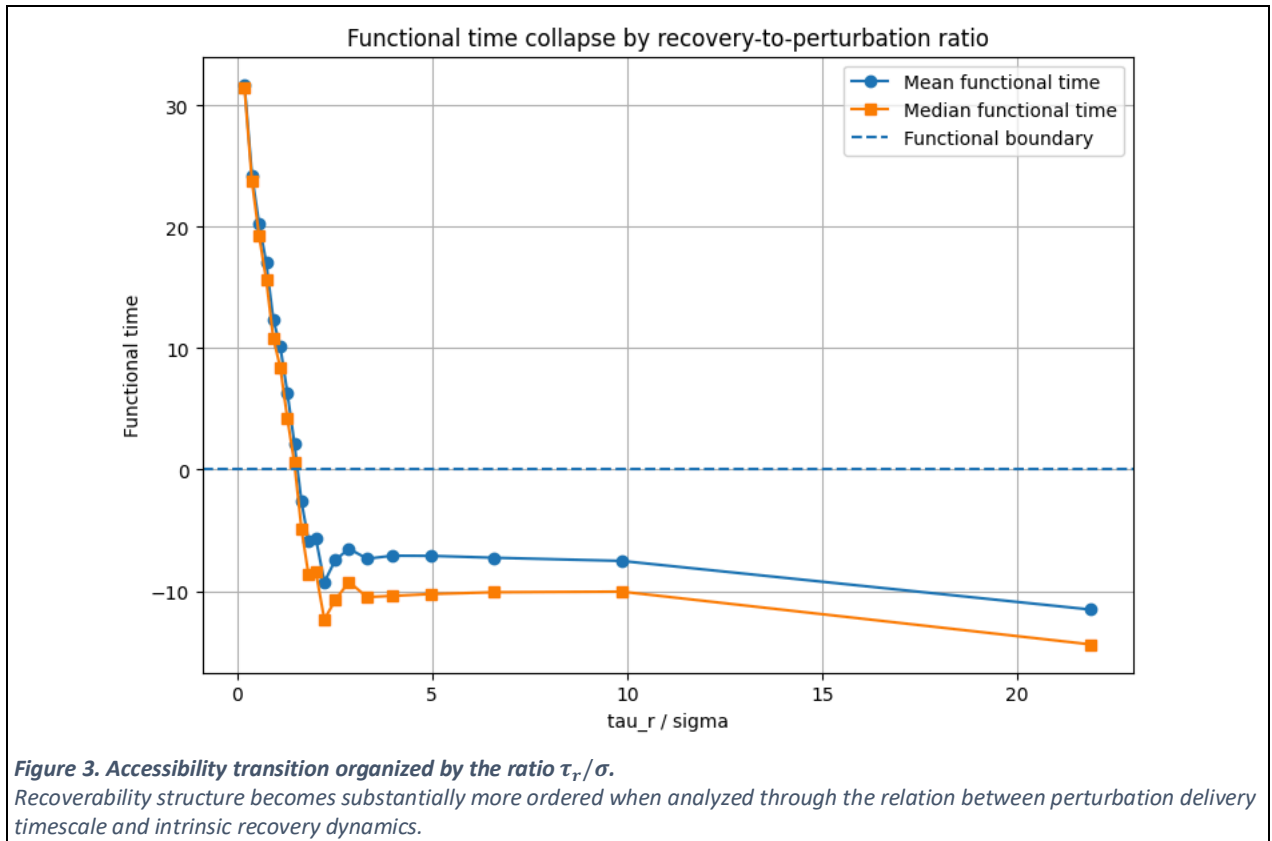
This ratio compares how rapidly perturbations are delivered relative to how rapidly the system can resolve and recover from them.

When recoverability was plotted against this ratio, the accessibility transition collapsed into a much more organized structure than when analyzed using perturbation magnitude alone. Systems with small τ_r / σ ratios generally remained operationally recoverable, indicating that recovery dynamics were sufficiently fast relative to perturbation delivery. In contrast, larger ratios produced progressively reduced accessibility and eventual recoverability collapse.

Importantly, the transition was not perfectly binary. Instead, the simulations revealed an intermediate transition region characterized by rapidly changing operational accessibility.

This organization suggests that recoverability is governed less by perturbation magnitude in isolation and more by the temporal matching between perturbation delivery and intrinsic recovery dynamics.

The emergence of this ratio organization was observed consistently across deterministic simulations, stochastic forcing conditions, and sequential perturbation experiments.



6. Sequential Perturbations and History Dependence

The previous simulations evaluated recoverability under isolated perturbation events. However, many real dynamical systems experience repeated perturbations before complete recovery from prior events has occurred.

To explore this possibility, the framework was extended to sequential perturbation experiments involving multiple temporally separated forcing events.

The simulations introduced an initial perturbation followed by a second perturbation after a controlled inter-event delay. Recoverability after the second perturbation was then evaluated using the same operational criterion applied throughout the framework.

The results showed that recoverability depended not only on the current perturbation, but also on unresolved prior dynamical state.

Systems that had not sufficiently recovered from the initial perturbation exhibited reduced accessibility during subsequent perturbation events, even when the later forcing itself remained unchanged.

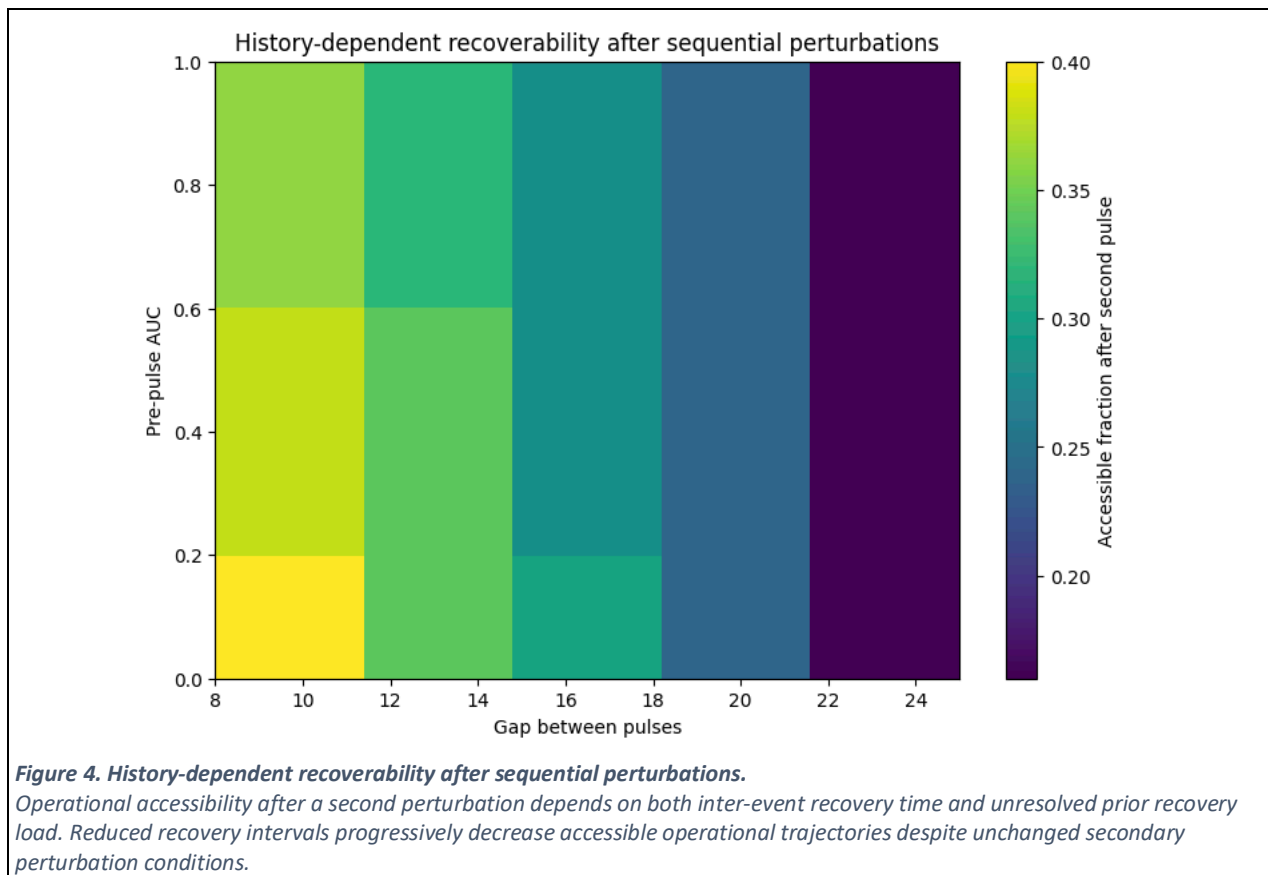
This generated a form of history-dependent recoverability in which operational accessibility became conditioned by prior unresolved recovery dynamics.

Importantly, the effect emerged without introducing explicit long-term memory variables or adaptive mechanisms. The history dependence arose naturally from finite recovery dynamics interacting with sequential perturbation timing.

The resulting transition landscape suggests that recoverability may depend on accumulated unresolved recovery load rather than perturbation magnitude alone.

These observations further support the distinction between:

- structural admissibility of trajectories,
- and operational accessibility of recoverable return pathways under finite-time constraints.



7. Stochastic Robustness

The previous analyses were performed under deterministic forcing conditions. To evaluate whether the observed recoverability structure persisted under more realistic variability, stochastic perturbations were introduced into the system dynamics.

Noise was incorporated using stochastic forcing terms added to the perturbation dynamics while preserving the same overall operational framework and recoverability criterion.

The objective was not to reproduce specific environmental noise statistics, but to test whether the recoverability boundary represented a fragile numerical artifact or a persistent dynamical structure.

Under stochastic forcing, the sharp deterministic recoverability boundary broadened into a probabilistic transition region.

Rather than producing a binary separation between recoverable and non-recoverable states, the system exhibited graded recovery probabilities across the transition zone.

Importantly, however, the overall accessibility structure persisted.

Regions that remained highly recoverable under deterministic forcing generally retained high recovery probability under stochastic conditions, while inaccessible regions remained operationally collapsed. The primary effect of noise was therefore to smooth the transition and transform it into a probabilistic accessibility landscape rather than eliminate it.

Additional simulations quantified variability in remaining functional recovery margin across parameter space.

Near the transition region, functional-margin variability increased substantially, indicating heightened sensitivity of operational accessibility close to the recoverability boundary.

This behavior suggests that recoverability transitions under stochastic forcing may naturally generate probabilistic accessibility landscapes rather than rigid deterministic thresholds.

The persistence of the transition structure under stochastic forcing substantially strengthens the interpretation that the observed recoverability organization reflects a genuine dynamical property of the system rather than a fragile deterministic artifact.

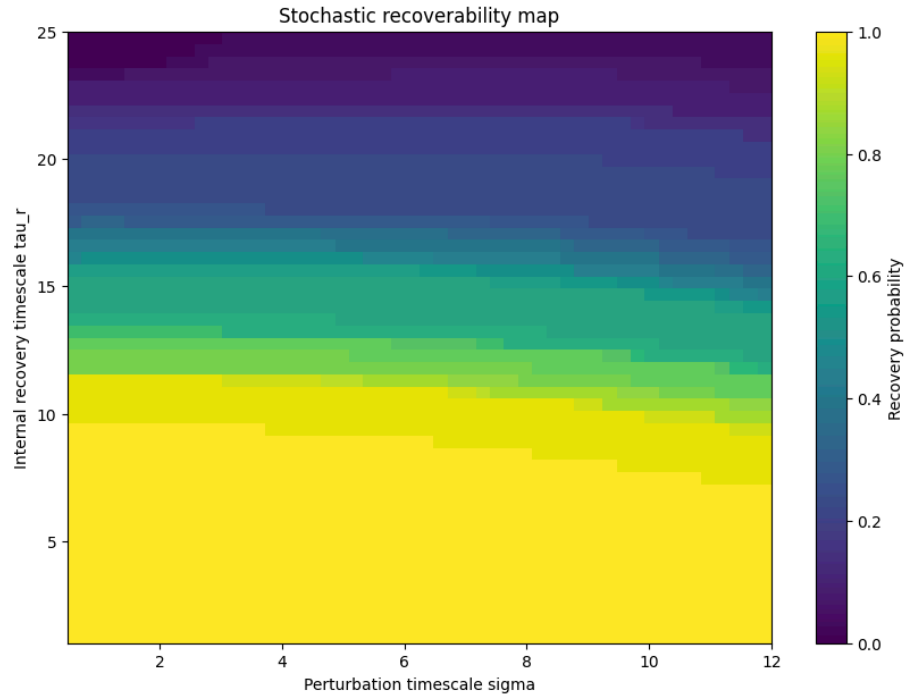


Figure 5. Stochastic forcing transforms the deterministic recoverability boundary into a probabilistic accessibility transition band while preserving the overall recoverability structure.

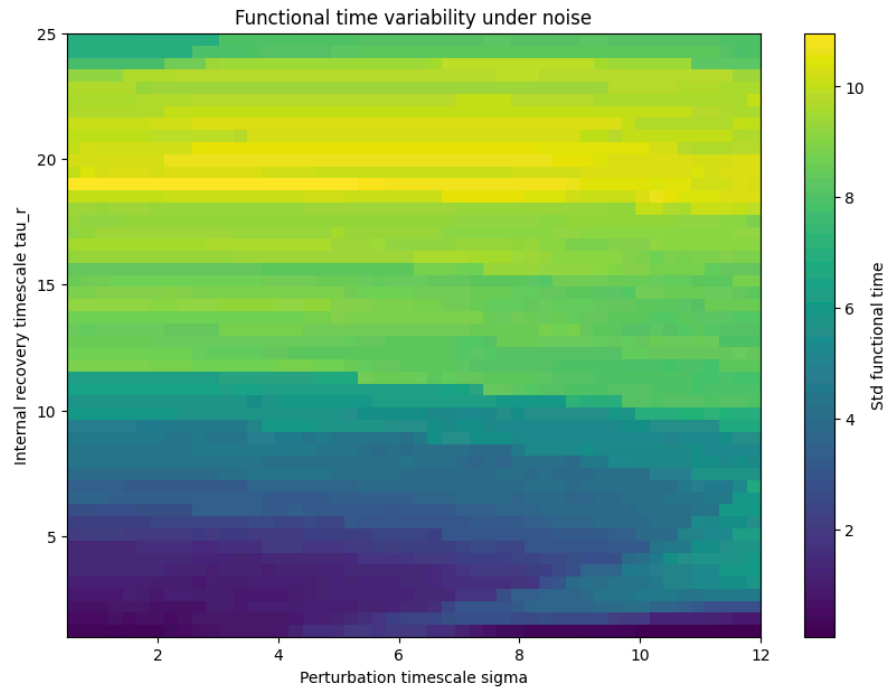


Figure 6. Variability in functional recovery margin increases near the transition region under stochastic forcing, indicating heightened sensitivity close to the recoverability boundary.

8. Numerical Robustness

To evaluate whether the observed recoverability structure depended strongly on numerical implementation details, additional simulations were performed across multiple numerical configurations.

The sensitivity analysis explored variations in:

- simulation timestep (DT),
- total simulation duration (T_MAX),
- and parameter-space sampling resolution.

The objective was to determine whether the recoverability boundary, functional-time landscape, and ratio organization remained qualitatively stable across numerical conditions.

The results showed that the primary operational accessibility structure persisted consistently across the tested numerical configurations.

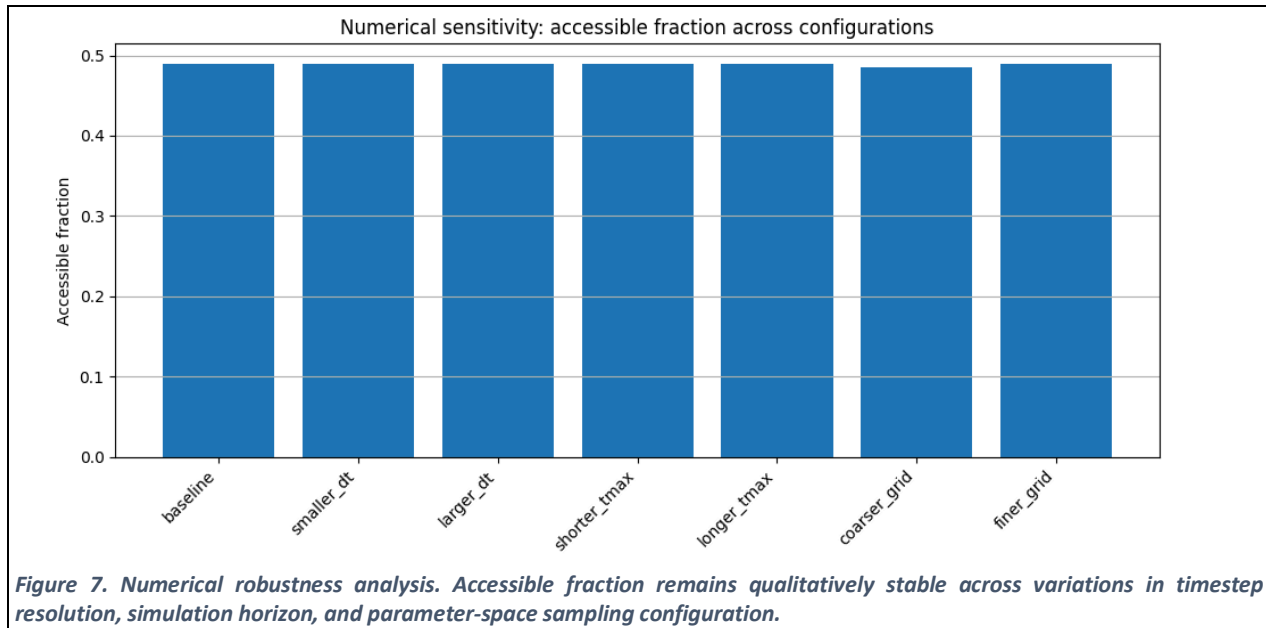
Although small quantitative shifts occurred near transition regions, the overall organization of:

- recoverable regions,
- inaccessible regions,
- ratio-dependent transitions,
- and stochastic accessibility bands remained stable.

Importantly, the accessibility transition did not disappear or qualitatively reorganize under finer timestep resolution or extended simulation duration.

This robustness substantially reduces the likelihood that the observed recoverability boundaries arise from discretization artifacts or insufficient simulation horizon.

The persistence of the main accessibility structure across deterministic, stochastic, sequential, and numerical sensitivity experiments supports the interpretation that the framework captures a stable operational property of recoverability-limited dynamical systems.



9. Interpretation and Limitations

The simulations presented here suggest that recoverability-limited systems may exhibit structured operational accessibility transitions governed by the interaction between perturbation delivery and intrinsic recovery dynamics.

Across deterministic, sequential, stochastic, and numerical sensitivity experiments, the primary observation remained consistent:

recoverability was not determined by perturbation magnitude alone.

Instead, operational accessibility depended strongly on the temporal relation between perturbation delivery and intrinsic recovery timescales.

The framework further suggests that systems may progressively lose operational recovery margin prior to complete loss of recoverability. In this sense, accessibility degradation can emerge continuously before full dynamical collapse occurs.

Importantly, the observed transitions do not imply disappearance of the underlying dynamical state space itself. Rather, the results suggest a distinction between:

- structural admissibility of trajectories,
- and operational accessibility of recoverable trajectories within finite functional time constraints.

The framework therefore remains operational rather than fundamental.

The present work does not propose modifications to physical time, relativity, spacetime geometry, or fundamental physical laws. The concept of “functional time” used throughout the notebook refers only to remaining operational recovery accessibility within finite-time dynamical constraints.

Similarly, the simulations do not attempt to model specific biological, cognitive, or physical mechanisms directly. The objective is instead to explore whether minimal recoverability-limited systems naturally generate organized accessibility transitions under finite-time forcing conditions.

Several important limitations remain.

First, the model is intentionally highly simplified and low-dimensional. Real systems may contain additional nonlinearities, adaptive responses, spatial coupling, heterogeneous recovery processes, and external control mechanisms not represented here.

Second, the operational recoverability criterion depends on finite observation windows and predefined recovery thresholds. Different operational definitions may alter quantitative transition boundaries.

Third, the current framework explores only minimal perturbation-recovery dynamics and does not establish universality across physical, biological, or computational systems.

Despite these limitations, the persistence of the observed accessibility structure across deterministic, stochastic, sequential, and numerical sensitivity experiments suggests that the framework may capture a useful minimal operational distinction between admissibility and recoverability under finite-time constraints.

10. Conclusion

This work introduced a minimal recoverability-limited dynamical framework for exploring operational accessibility under finite-time perturbation and recovery constraints.

The simulations consistently showed that recoverability is not determined solely by perturbation magnitude. Instead, operational accessibility depends strongly on the relation between perturbation delivery timescale and intrinsic recovery dynamics.

Across deterministic, sequential, stochastic, and numerical sensitivity experiments, the framework generated:

- structured recoverability boundaries,
- gradual degradation of functional recovery margin,
- ratio-organized accessibility transitions,

- history-dependent recoverability,
- and probabilistic transition regions under stochastic forcing.

Importantly, the observed transitions emerged while the underlying dynamical state space remained structurally admissible.

The results therefore support a minimal operational distinction between:

- existence of possible trajectories,
- and finite-time accessibility of recoverable trajectories.

Within this framework, systems may lose recoverable operational accessibility before complete dynamical collapse occurs.

The present study is exploratory and intentionally minimal. The objective is not to propose a general theory of time or universal physical mechanism, but to provide a reproducible computational framework for studying recoverability-limited operational accessibility under finite-time dynamical constraints.

Future work may explore whether similar accessibility structures emerge in higher-dimensional, spatially distributed, adaptive, or experimentally measurable systems.

11. Appendix

Appendix A — Parameters and Reproducibility

All simulations, figures, and tables presented in this document were generated directly from the accompanying reproducible computational notebook.

The notebook contains:

- the complete simulation framework,
- deterministic and stochastic simulations,
- sequential perturbation experiments,
- numerical sensitivity analyses,
- figure-generation routines,
- parameter tables,
- and exported result summaries.

The framework uses only openly available computational dependencies and does not require proprietary datasets or external private resources.

Core simulation parameters included:

- perturbation load (AUC),
- perturbation delivery timescale (σ),
- intrinsic recovery timescale (τ_r),
- functional recovery window (T_window),
- stochastic forcing amplitude,
- numerical timestep resolution (DT),
- and simulation horizon (T_MAX).

All numerical values used throughout the simulations are included directly in the notebook outputs and exported parameter tables.

Appendix B — Repository and Reproducibility

Notebook:

Functional_Time_ERC.ipynb

Figures:

All figures were generated directly from the notebook simulations.

Reproducibility:

The complete computational notebook contains all simulations, parameter sweeps, figure-generation routines, and exported outputs required to reproduce the results presented in this document.

12. References

- Ashwin, P., Wieczorek, S., Vitolo, R., & Cox, P. (2012).
Tipping points in open systems: bifurcation, noise-induced and rate-dependent examples in the climate system.
Philosophical Transactions of the Royal Society A, 370(1962), 1166–1184.
- Ritchie, P., & Sieber, J. (2016).
Early-warning indicators for rate-induced tipping.
Chaos: An Interdisciplinary Journal of Nonlinear Science, 26(9), 093116.
- Wieczorek, S. (2023).
Rate-induced tipping and transient dynamics in nonequilibrium systems.
Annual Review of Condensed Matter Physics, 14, 413–436.

- Scheffer, M., Bascompte, J., Brock, W.A., Brovkin, V., Carpenter, S.R., Dakos, V., Held, H., van Nes, E.H., Rietkerk, M., & Sugihara, G. (2009).
Early-warning signals for critical transitions.
Nature, 461, 53–59.
- Holling, C.S. (1973).
Resilience and stability of ecological systems.
Annual Review of Ecology and Systematics, 4, 1–23.
- Ojeda, J.
Accessibility vs Admissibility in Recoverability-Limited Dynamical Systems.
Zenodo. doi: 10.5281/zenodo.19412131
- Ojeda, J.
Endogenous Reachability Collapse (ERC): Minimal models of recoverability boundaries.
Zenodo. doi: 10.5281/zenodo.19240820